北京郵電大學

Queen Mary University of London

For examiners' use only

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EBU6018 A

Joint Programme Examinations 2021/22

EBU6018

Paper A

Time allowed 2 hours

Answer ALL questions

Complete the information below about yourself very carefully.

QM student number										
BUPT student number										
Class number										

Electronic calculators are allowed but NOT electronic dictionaries.

INSTRUCTIONS

1. **You must NOT take answer books, used or unused, from the examination room.**
2. Write only with a black or blue pen **and in English**.
3. Do all rough work in the answer book – **do not tear out any pages**.
4. If you use Supplementary Answer Books, tie them to the end of this book.
5. Write clearly and legibly.
6. **Read the instructions on the inside cover.**

Examiners

Mr Andy Watson, Dr Cunhua PAN

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Instructions

Before the start of the examination

1) Place your BUPT and QM student cards on the corner of your desk so that your picture is visible.

2) Put all bags, coats and other belongings at the back/front of the room. All small items in your pockets, including wallets, mobile phones and other electronic devices must be **placed in your bag in advance**. **Possession of mobile phones, electronic devices and unauthorised materials is an offence.**

3) Please ensure your mobile phone is switched off and that no alarm will sound during the exam. **A mobile phone causing a disruption is also an assessment offence.**

4) Do not turn over your question paper or begin writing until told to do.

During the examination

1) You must not communicate with or copy from another student.

2) If you require any assistance or wish to leave the examination room for any reason, please raise your hand to attract the attention of the invigilator.

3) If you finish the examination early you may leave, but not in the first 30 minutes or the last 10 minutes.

4) For 2 hour examinations you may **not** leave temporarily.

5) For examinations longer than 2 hours you **may** leave temporarily but not in the first 2 hours or the last 30 minutes.

At the end of the examination

1) You must stop writing immediately – **if you continue writing after being told to stop, that is an assessment offence.**

2) Remain in your seat until you are told you may leave.

Question 1

a) A set of 3 mutually orthogonal vectors can be used as a BASIS for R^3.

Matrix $A = \begin{bmatrix} 2 & -2 & x \\ -3 & -1 & y \\ 1 & 1 & z \end{bmatrix}$ **[16 marks]**

i) Find the values of x, y and z so that the columns of A form an orthogonal basis for R^3. **(9 marks)**

ii) If vector $u = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$, find c_1 , c_2 and c_3 so that $x = c_1v_1 + c_2v_2 + c_3v_3$. **(7 marks)**

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	16 marks

b) The Fast Fourier Transform (FFT) is a method for reducing the time taken to perform a Discrete Fourier Transform (DFT). Using the given 8-point sequence, describe the process of implementing an FFT. Refer to radix-2 decimation in time.

[3, -4, 8, -2, -7, 9, 13, 6]

[9 marks]

	Do not write in this column

		9 marks

Question marking: $\frac{}{16} + \frac{}{9} = \frac{}{25}$

Question 2

a) The Discrete Fourier Transform can be defined as

$$X[k] = \sum_{n=0}^{N-1} x[n] W_N^{-nk}$$

where $W_N = e^{\frac{j2\pi}{N}}$ **[18 marks]**

An N-point DFT can be written as X = Wx, where x is the N-point input sequence of samples of a continuous signal, W is the N-by-N DFT matrix and X is the DFT of the signal

i) Show that the normalised 4x4 Fourier Matrix is:

$$F_4 = \frac{1}{2}\begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & -i & -1 & i \\ 1 & -1 & 1 & -1 \\ 1 & i & -1 & -i \end{bmatrix}$$ **(14 marks)**

ii) Hence, calculate the DFT of the 4-point input sequence x= [3, 0, 4, 2]. **(4 marks)**

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	18 marks

b) This question is about the Fourier Transform. **[7 marks]**

i) Explain the limitation of the Fourier Transform in the context of non-stationary signals.

(2 marks)

ii) Explain how a Short-Time Fourier Transform overcomes the limitation of the Fourier Transform and state two limitations of the Short-Time Fourier Transform.

(5 marks)

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		7 marks

Question marking: $\frac{}{18} + \frac{}{7} = \frac{}{25}$

Question 3

This question is about the Karhunen-Loeve Transform (KLT). **[25 marks]**

a) List the steps in performing a KLT. **(13 marks)**

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		13 marks

b) Table Q3 b) shows a 2D data set which contains a sample of a larger set.

x	y
2.5	2.4
0.5	0.7
2.2	2.9
1.9	2.2
3.1	3.0
2.3	2.7
2.0	1.6
1.0	1.1
1.5	1.6
1.1	0.9

Table Q3 b)

Show that the covariance matrix is (to 4 decimal places):

$$cov_{x,y} = \begin{bmatrix} 0.6166 & 0.6154 \\ 0.6154 & 0.7166 \end{bmatrix}$$ **(12 marks)**

NOTE: you must show all your working.

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		12 marks

Question marking: $\frac{}{13} + \frac{}{12} = \frac{}{25}$

Question 4

This question is about wavelet transforms using filterbanks. **[25 marks]**

a) In the wavelet transform, the scaling function coefficients $c_{m,n}$ and wavelet series coefficients $d_{m,n}$ can be calculated recursively according to the following equations:

$$c_{m-1,n} = \sqrt{2}\sum_i h_0[i-2n]c_{m,i}$$
$$d_{m-1,n} = \sqrt{2}\sum_i h_1[i-2n]c_{m,i}$$

Explain how this can be interpreted in terms of filtering and downsampling, and hence leads to the concept of an *analysis filterbank*. Sketch a diagram to illustrate this filterbank.

(6 marks)

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		6 marks

b) Suppose we have a Haar wavelet transform analysis filterbank which uses a low-pass filter

$h_0[0] = h_0[1] = \frac{1}{2}$ and a high-pass filter $h_1[0] = \frac{1}{2}$, $h_1[1] = -\frac{1}{2}$.

Use the recursive equations:

$$c_{m-1,n} = \sqrt{2} \cdot \frac{1}{2}(c_{m,2n} + c_{m,2n+1})$$
$$= \frac{1}{\sqrt{2}}(c_{m,2n} + c_{m,2n+1})$$
$$d_{m-1,n} = \frac{1}{\sqrt{2}}(c_{m,2n} - c_{m,2n+1})$$

to calculate the Haar wavelet transform for a sampled signal s[n] = [4, 7, 6, -2] after 1 and 2 stages of the transform filterbank.

(7 marks)

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	7 marks

c) With reference to the general filterbank block diagram shown in Figure Q4 c), explain what is meant by the term "Perfect Reconstruction" and how this helps to design the filters.

(6 marks)

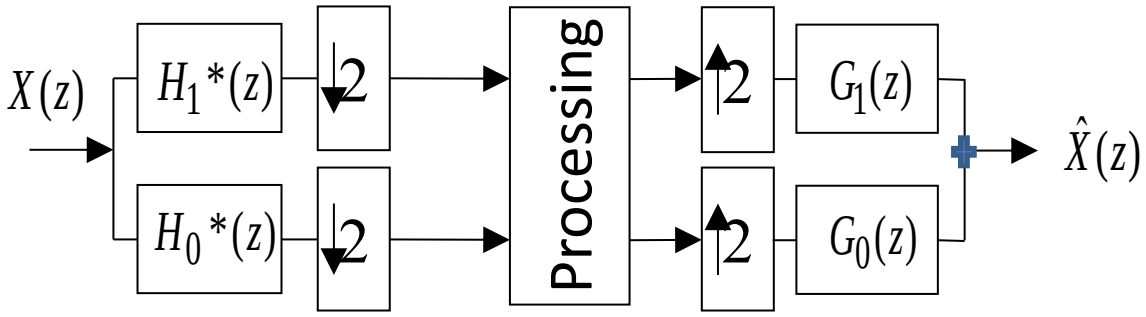


Figure Q4 c)

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		6 marks

d) Filter banks are used to implement wavelet transforms.

In the diagram of Q4c), the analysis low pass filter is referred to as H_0 and the high pass filter as H_1. **T**he synthesis low pass filter is referred to as G_0 and the high pass filter as G_1

For orthogonal analysis filters:

$$H_1(z) = (-z)^{-N} H_0(-z^{-1})$$

And for the synthesis filters

$$G_0(z) = H_1(-z)$$

$$G_1(z) = -H_0(-z)$$

Daubechies wavelets are orthogonal. For a Daubechies 2nd order wavelet,

$h_0[n]$ is defined by the sequence [0.483, 0.837, 0.224, -0.129].

Determine $h_1[n]$, $\gamma_0[n]$ and $\gamma_1[n]$.

(6 marks)

	Do not write in this column

		6 marks

Question marking: $\frac{}{6} + \frac{}{6} + \frac{}{7} + \frac{}{6} = \frac{}{25}$

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